

OPEN SOURCE · CANADA (US EXPERIMENTAL) · RUNS ON YOUR MACHINE

taxjson

Your broker's CSV in. Schedule 3 out. Capital gains, dividends and the superficial-loss rule computed the way the CRA defines them — across every account you own — with every rule citing its source.

```
bash -c "$(curl -fsSL https://taxjson.com/install.sh)"
```

Installs `taxjson` and its short name `tjs` · taxjson.com · github.com/taxjson/taxjson · MIT

THE PROBLEM

The slip your broker files is not your cost basis

A T5008 reports proceeds per account. The Income Tax Act taxes you on **one average cost per security across all your accounts**, denies losses when any affiliated account — your RRSP included — bought the same thing within 30 days, and folds option premiums, splits, spin-offs and return of capital into that cost. Nobody's statement does that math for you.

Cross-account pooling

s.47: one ACB pool per identical property. A US listing and its TSX listing are the same pool; a CDR is not.

Superficial losses

s.54 / s.40(2)(g): a loss is denied if you *or an affiliated person* bought within ± 30 days and still holds at day 30 — sized sale by sale. Denied into an RRSP, it is gone for good.

Everything that moves cost

Assignments (IT-479R), warrant exercises, splits, renames, spin-offs (s. 86.1), mergers with elections, ROC, reinvested distributions — in the right order, on the settlement date.

Spreadsheets get the easy 90% right. The other 10% is where reassessments come from.

WHAT IT DOES

Three steps, all on disk

inputs/

Broker CSV exports per account: IBKR, Questrade, RBC, Webull, Kraken, Coinbase — plus plain-text `.tt` for anything else



tjs run

Parse · convert at the Bank of Canada rate per settlement date · pool identical securities · dated renames and corporate actions with elections · superficial-loss pass across taxable and registered accounts



reports/

Per-account summaries, Schedule 3 / Form 8949 rows, the wash radar, T1135, a tax and minimum-tax estimate — and the broker cross-check

Two project files

`taxjson.toml` : year, country, accounts (taxable / sheltered) and the figures you enter by hand. `ticker.map` : dated renames and symbol lookups.
`tjs migrate` converts an older project.

Row accounting

Every parser accounts for every row it reads. Rows it recognises but does not translate are counted and named — never silently dropped.

Traceable numbers

`taxjson audit` recomputes every disposition from the parsed broker row through FX, ACB and the wash determination, and ties it out against the reports.

Every computation names its source

RULE	WHAT TAXJSON DOES	SOURCE
Identical properties	One average-cost pool per security across accounts; cross-listings merged, CDRs kept apart; a ticker change carries the pool on its date	s.47(1); IT-387R2; T4037
Superficial loss	CRA's formula, sale by sale: the least of the units sold, bought in the 61-day window, and held at day 30; the RRSP/TFSA is affiliated; denied-into-plan is permanent	s.54, s.40(2)(g)(i), s.53(1)(f), s.251.1(1)(g)
Timing	Sales on the settlement date, including at year end; a dividend in the year paid, a Canadian trust's distribution in the year payable	T4037; s.82(1); s.104(13)
Foreign currency	Bank of Canada rate on each settlement date, never an annual average	Folio S5-F4-C1
Options	A written option's premium is a gain in the year written; a buy-back is a loss in its own year; an assignment folds the premium into the shares; an exercised warrant's cost goes into the shares too. <code>option-boundary</code> says when a filed year needs a T1-ADJ	s.49(1)–(4); IT-479R
Corporate actions	Splits, spin-offs, mergers with the rollover election recorded and a filing reminder at every run; ROC reduces ACB	s.86.1, s.85.1, s.53(2)(h)
US (experimental)	FIFO; §1091 wash sales with basis and holding-period carry, IRA repurchases permanent; lots carried between your own accounts; §355 / §356 per lot; Form 8949 codes	§1091, Rev. Rul. 2008-5; Reg. 1.358-2, 1.356-1(b)

`tjs tax-logic` prints every rule taxjson applies, one line each, with the id its tests cite — Canada's and the USA's apart, never mixed. REFERENCES.md maps each to its source, including what is deliberately *not* modelled. A rule that disagrees with its source is a bug; that is the most useful issue the project can receive.

WORKED EXAMPLE

A loss you cannot claim

What happened

The margin account sells 400 shares at a loss on February 3. On February 10 the RRSP buys 200 of the same stock, and still holds them on March 5.

What the slip says

A capital loss on 400 shares. The spreadsheet agrees.

What the Act says

CRA's formula: the least of 400 sold, 200 acquired in the 61-day window and 200 still held at its end — the loss on 200 shares is superficial (s.54). The ACB bump goes to the RRSP, where it is worthless: permanently denied. Shares the RRSP held before the window would not have counted.

NOTE: superficial loss – 200 sh denied (permanently: replacement property held in rrsp)
loss sale 2026-02-03 · trigger lot 2026-02-10 (rrsp) · window end 2026-03-05

Only a tool that sees every account at once can get this right. That is the whole design.

Checked against the one thing that cannot lie

Internal reports can all agree and still be wrong. `taxjson sanity` compares the positions the books say you hold with the positions your broker's own export says you hold, account by account. Any difference is a trade that has not reached the CSV yet, or a booking problem — there is no third kind.

6,500+ tests

Every fix lands with a regression test that failed before it, tagged with the tax-logic rule it pins; the gate fails on a rule with no test.

Fuzzers and mutants

Three property fuzzers run thousands of random ledgers per run; mutation testing of both engines finds the boundaries no test pins.

Two full audits

About 1,300 and then 1,600 findings, after eight review rounds and a security audit. Every confirmed finding reproduced, fixed, pinned.

Published limits

KNOWN_ISSUES.md lists what is out of scope and what to do by hand, with the citation.

COMMANDS

One pipeline, a few dozen views on it

Build the books

- `init` a new project
- `migrate` an older one
- `fetch` from your broker (plugin)
- `run` the pipeline + broker check
- `elect` corporate actions

Read the numbers

- `sum` the year, by form line
- `estimate` / `amt` the tax
- `stats` wins and losses
- `list` / `shares` positions
- `events` / `trades` rows

Before you trade

- `wash-radar` open windows
- `buy-check` / `sell-check`
- `harvest` losses to take
- `scan` tax-efficiency lint

Before you file

- `checklist` every step
- `form-export` Schedule 3
- `reconcile-slips` T5008
- `t1135` / `carryover`
- `close-year` / `handoff`

Explain

- `audit` every gain
- `tax-logic` every rule
- `renames` dated events
- `sanity` vs the broker
- `edge-cases` year ends

`tjs` with no command lists them by purpose, one line each — inside a project only your country's. Every command takes `-h`; most take `--json`. `tjs redact` strips identifiers from a real export so it can be shared as a parser sample.

Nothing personal leaves the machine

Local by construction

Parsing, pooling and the wash pass run on your computer — no server, no web UI, just commands and files. The pipeline's only network calls are FX rates (Bank of Canada, its noon rate before March 2017; Yahoo only before May 2007) and crypto prices for unpriced rows — both switchable off with `TAXJSON_OFFLINE=1`. SECURITY.md lists every egress.

No credentials in the core

The core holds no broker client and reads no credential. Broker fetch is a separate, optional package (taxjson-fetch): it uses your own token file, never writes it into a project, and sends it nowhere but that broker.

The project guards itself

A personal-data scan runs in every gate and as the pre-push hook, fails closed, and reads a private denylist kept outside the repository; the hook also refuses a commit message that quotes a money total. `redact` keeps every row shape and drops the identifiers it recognises — review before sharing.

Zero runtime dependencies in the core. A tax tool has to be trustworthy before it is clever.

SCOPE

Canada is the product. The US engine is a draft.

Canada — supported

Average-cost ACB, the superficial-loss rule across accounts, settlement-date timing, options and warrants, dated renames, corporate actions with elections, T1135, Schedule 3, a marginal-rate estimate with provincial tables, minimum tax line by line, instalments, and carry-forwards from one year's lock to the next. Validated against two years of real multi-account books and the brokers' positions.

US — experimental

FIFO, §1091 wash sales across your accounts, lots carried between your own accounts, §355 / §356 per lot, short/long-term, Form 8949 codes and a TXF export. Implemented and unit-tested, never validated on a real account. It says so at every run. Real US exports — redacted — are the way to change that.

The two rule sets never mix: a project has one country, and a setting, flag or command the other country owns is refused, not silently applied.

Honest labelling costs nothing. A tax tool that overclaims loses the only thing it has.

BROKERS

Reads the exports you already have

BROKER	FORMAT	NOTES
Interactive Brokers	Activity / Flex statement CSV	Trades, options, corporate actions, warrant exercises, transfers, fees, dividends; Flex auto-fetch (taxjson-fetch plugin)
Questrade	Activity CSV	Trades, DRIP, cash-in-lieu, transfers; REST auto-fetch and live positions (taxjson-fetch plugin)
RBC Direct Investing	Activity export	Trades, dividends with withholding, warrant exercises
Webull	Trading Summary CSV	Trades and options (BUY/SELL rows only); no income — T5 interest and dividends go in as <code>.tt</code> lines
Kraken · Coinbase	Ledgers / transaction history	Trades, rewards, sends and withdrawals as custody evidence
Anything else	<code>.tt</code> plain text · generic CSV mapping	Ten-line files a human can write and read

Help wanted: Wealthsimple, TD Direct, BMO InvestorLine, CIBC Investor's Edge, Scotia iTRADE, National Bank Direct, Qtrade, Desjardins. A parser is an afternoon's work given one real export — `tjs` `redact` makes sharing one safe. An auto-fetcher for another broker is a plugin of its own.

INSTALL

Your first books in ten minutes

```
bash -c "$(curl -fsSL https://taxjson.com/install.sh)"
```

Installs the latest release into its own Python environment and links `taxjson` and its short name `tjs` onto your PATH. Needs git and Python 3.9+, nothing else. Re-run to upgrade; add `--with-fetch` for the Questrade / IBKR auto-fetch plugin.

```
mkdir -p ~/taxes/2026 && cd ~/taxes/2026
tjs init --country canada      # taxjson.toml, ticker.map, inputs/
# drop broker CSVs into inputs/margin/, inputs/rrsp/, ...
tjs run                        # books, reports, wash pass, broker check
tjs sum                        # the year, by Schedule 3 line
tjs wash-radar                 # "can I sell this at a loss today?"
tjs checklist                  # at year end: ready to file?
```

WHERE IT GOES

Roadmap

More Canadian brokers

Driven by redacted real exports from users. One parser per afternoon, each with the row-accounting contract; auto-fetch for more brokers as plugins.

Rules retail investors hit

Non-eligible dividends and T3 box splits, the s.39(4) election, CNIL — each stated in tax-logic with its tests, whichever real exports show first.

US validation

The engine exists; it needs a real account. The label comes off when the numbers have been checked against one.

1.0

PyPI, a frozen CLI, and the interchange contract with portfolio tools written down.

taxjson.com · github.com/taxjson/taxjson · MIT · not tax advice — a calculator that shows its work.